

AGRO-IoT SOLUTIONS

Technically Empower Indian Farmers



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India is fast becoming a hotbed for research on the Internet of Things (IoT), machine learning, artificial intelligence and more. The engineering and research community is optimistic that the country can do away with manual ways of doing things across various industries, speeding up the processes while also making them transparent.

One field that needs immediate intervention is agriculture and farming, where manual bureaucracy has denied farmers (more so marginal farmers) government-fixed prices for their produce.

How IoT and other smart concepts transform farming

"IoT is helpful in instantaneous collection of detailed data. The tracking of quantity, quality, time and price at various points of exchanges will help us to focus on areas of improvement," states Dr Regi Mathew, data scientist at CleverInsight—a firm that is focused on leveraging the Internet of Things to transform businesses across the world.

Dr Regi believes that the right IoT-driven data collection mechanism will allow farmers (and the agricultural sector as a whole) to actually monitor the amount of produce that enters wholesale retailers.

Currently, most farmers still prefer middlemen (and agents) to find sales channels for their produce. This leads to a scenario wherein agents become price fixers and buy

farmers' produce at prices that are generally way below the market rates. Often, agents and middlemen quote insufficient supply or inferior quality of supply. This creates an artificial scarcity situation with respect to quantity and quality, leading to profit loss for farmers.

Creativity of embedded engineers and IoT service providers comes into play here, with respect to developing sensors that can monitor the amount of produce along with the quality of the supply in wholesale trading. Here, farmers should be offered technical insights in a simple manner so that they can gauge the end-results (in terms of quality of their produce) and directly deliver these to a wholesale retailer.

Farmers can be offered insights via mobile apps, and even text messages, after analytics is run on the data collected by sensors. Wholesale traders can directly connect with farmers after going through the agricultural data analysis (organised based on location as well) and revert with the current market prices. Thus middlemen are eliminated from the picture altogether.

Big Data: A 'big' advantage for farmers

"Agriculture production and distribution is an ideal case of Big Data application," informs Dr Regi. This means that technologists should work on presenting the agriculturists with dashboards that are capable